

New to ML? Follow These Steps to Build Any Machine Learning Model

Introduction

The video aims to empower viewers with machine learning skills rather than fear-mongering about AI taking over jobs. It emphasizes the importance of acquiring ML skills gradually, which will make individuals irreplaceable in their roles, regardless of advancements in generative AI or large agents. The core of the video outlines generalized steps applicable to training and testing any machine learning algorithm, whether for regression, classification, clustering, or neural networks. These are mandatory steps for building ML models.

Example: Supervised Learning (Regression)

The presenter uses regression, a type of supervised learning, as a running example.

- **Concept:** In supervised learning, the model works with both independent (input) and dependent (output) variables.
- **Process:** The model is provided with data containing both the input variables and the corresponding target output variable.
- **Scenario:** Predicting house prices.
 - **Input Data:** Number of rooms, location, size, etc.
 - **Output Data (Target):** The known price of the house based on its features.
 - **Action:** The model is trained on this data and then tested to see how well it predicts prices for new data.

General Steps to Build a Machine Learning Model

1. Load Data

- The fundamental first step before applying any ML, AI, or Big Data model.
- You must have data to work with (e.g., the house price dataset).
- **Common Format:** Often loaded in CSV (Comma Separated Values) format.
- **Common Library (Python):** `pandas` is widely used for loading data.
 - **Code Example:**

```
import pandas as pd
data = pd.read_csv('data.csv')
```

2. Pre-process Data

- Real-world data is rarely perfect; it needs cleaning and formatting.

- **Handling Missing Values:**
 - Real data often has missing entries (e.g., a row where 'salary' is missing).
 - *Action:* You can choose to delete rows with missing values or fill them in (e.g., using the mean or median of the column).
 - **Code Example:** `data.dropna()`
- **Encoding Categorical Data:**
 - ML models require numerical data. Categorical data (like 'Gender: Male/Female' or 'City: Delhi/Mumbai/Chennai') must be converted.
 - *Action:* Convert text labels into numbers (e.g., Male=0, Female=1; Delhi=0, Mumbai=1, Chennai=2).
 - **Code Example:** `pd.get_dummies(data)`
- **Feature Scaling:**
 - Ensures all features are on a similar scale so no single feature dominates the model (e.g., Age ranges from 20-60, while Salary ranges from 50,000-100,000).
 - *Action:* Normalize or standardize data to bring values within a similar range (e.g., 0 to 1).
 - **Code Example:**

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)
```

3. Split Data

- It's crucial not to train the model on the entire dataset. You need unseen data to evaluate its actual performance.
- **Train/Test Split:** Divide the data into a training set and a testing set.
- **Common Ratio:** 80% for training, 20% for testing.
- **Why?** If you train and test on the same data, the model might just memorize the answers (overfitting) and perform poorly on new data.
- **Code Example:**

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

4. Choose a Model

- Select the appropriate ML algorithm based on your problem (e.g., regression for continuous values, classification for categories).
- **Example (Regression):** Linear Regression.

- **Code Example:**

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
```

5. Train Model

- Feed the training data (`X_train` inputs and `y_train` outputs) to the chosen algorithm.
- The model learns the relationship between the inputs and the output.
- **Code Example:** `model.fit(X_train, y_train)`

6. Make Predictions

- Use the trained model to predict outcomes on the unseen test data (`X_test`).
- **Code Example:** `predictions = model.predict(X_test)`

7. Evaluate Model

- Compare the model's predictions against the actual true values (`y_test`) from the testing set to see how well it performed.
- **Evaluation Metrics:** Different metrics are used depending on the problem (e.g., Mean Absolute Error for regression).
 - **Mean Absolute Error (MAE):** Calculates the average difference between the predicted values and the actual values.
- **Goal:** Minimize the error.
- **Code Example:**

```
from sklearn.metrics import mean_absolute_error
error = mean_absolute_error(y_test, predictions)
```

Conclusion & Next Steps

- These are the generalized, mandatory steps for building an ML model.
- **Recommendation:** Start by applying these steps to small datasets using various algorithms (e.g., Linear Regression, SVM, Decision Trees, Neural Networks) to gain hands-on practice before moving to large datasets.